

AI/ML Hackathon – Ministry of Panchayati Raj (Geospatial Intelligence Challenge)

*Powered by: **Geo-Intel Lab, IITNiF***

Problem Statement Addressed

AI Based Feature Extraction from Drone Images

AI-Powered Geospatial Mapping of Rural Infrastructure Using Drone Orthophotos

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Title of the Idea:

AI-Powered Geospatial Mapping of Rural Infrastructure Using Drone Orthophotos.

Team & Affiliation:

Institution: Sagi Ramakrishnam Raju Engineering college

Team:

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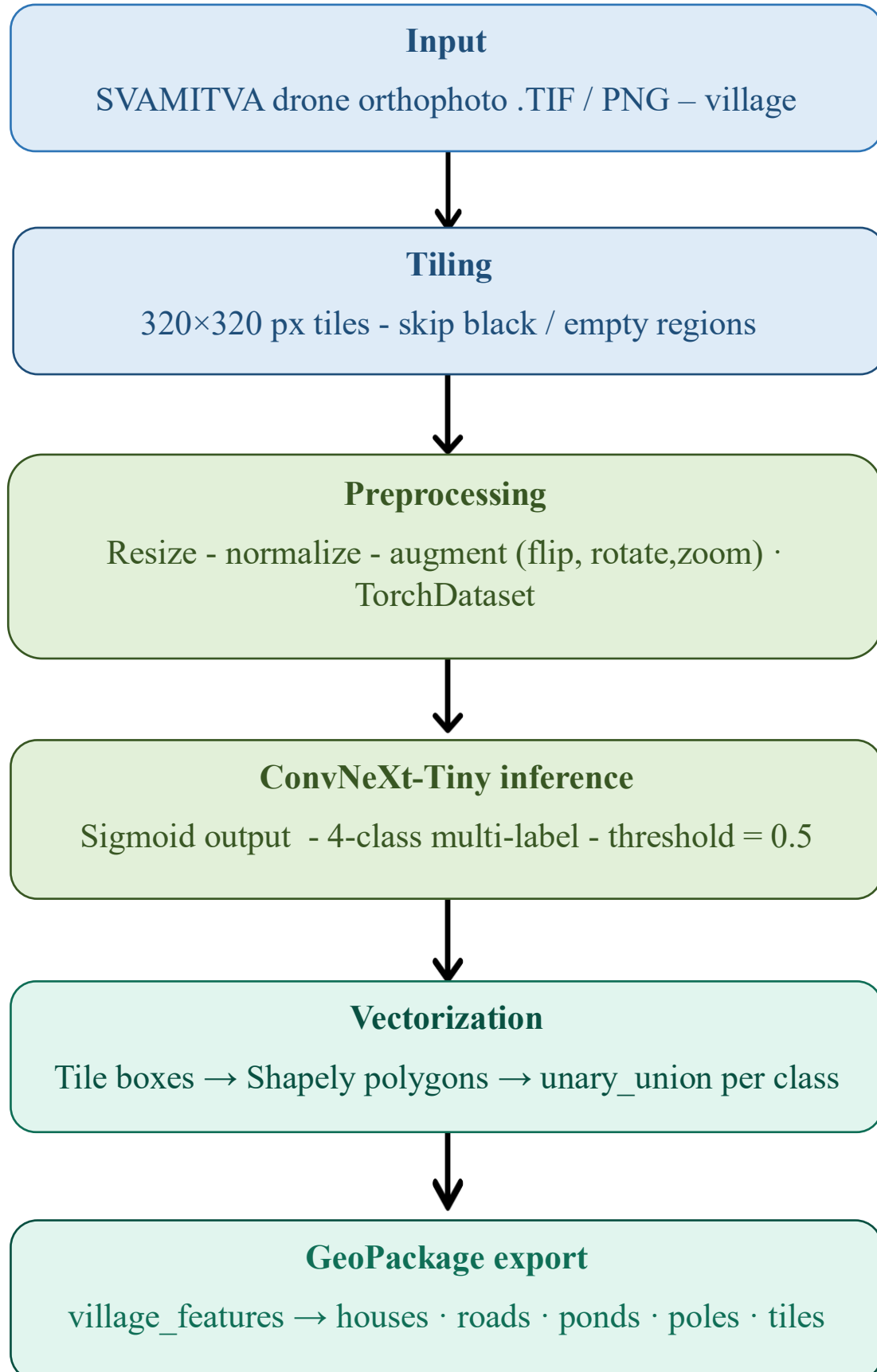
Mentor:

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Problem Statement Addressed:

Rural areas in India often lack updated maps showing roads, buildings, and other infrastructure. Traditional methods like field surveys take too much time and money to do regularly. As a result, local government bodies like Panchayati Raj institutions often make planning decisions without accurate data. This project solves that problem by using AI to automatically read Drone images and find infrastructure features. The system then turns this information into digital maps that can be used right away. This saves time, reduces cost, and gives officials reliable, up-to-date data for better rural development planning.

Proposed Solution:



Uniqueness & Innovation:

1. Advanced Architecture:

- Uses **ConvNeXt-Tiny** CNN backbone
- Pretrained on ImageNet for better aerial image performance

2. Multi-Label Classification :

- Each 320×320 tile can detect multiple features
- Identifies **Houses, Roads, Ponds** and **Electric Poles**
- Uses **sigmoid output** for overlapping classes

3. Class Imbalance Handling:

- Uses weighted loss (BCEWithLogitsLoss)
- Gives higher importance to rare classes
- Improves detection of features like electric poles

4. Tile-Based Inference:

- Large village image → split into tiles
- Enables **scalable processing** for any image size

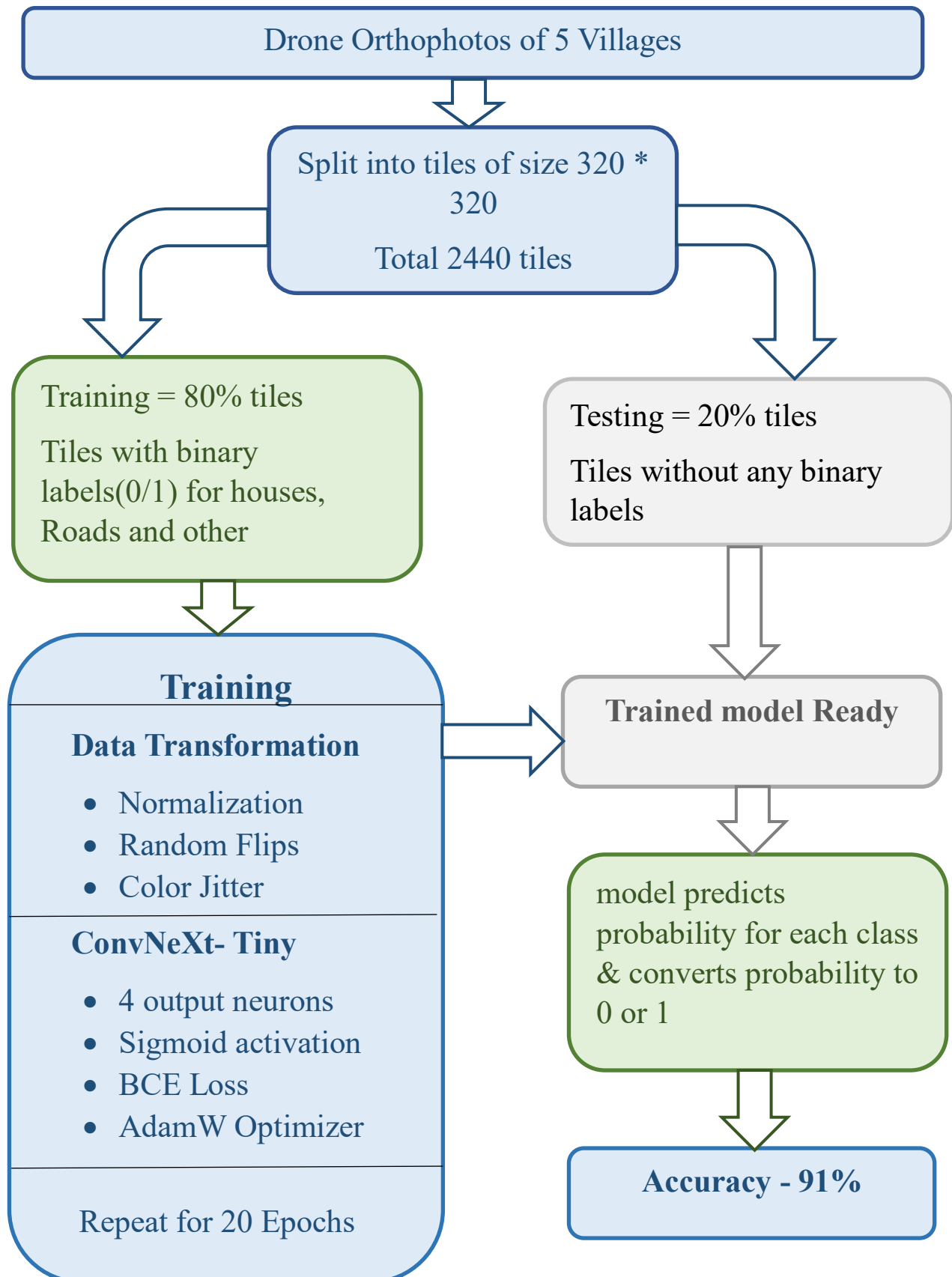
5. End-to-End Automation:

- Single pipeline: Image → Prediction → Geo-output
- No need for manual annotation or GIS experts

Technology Stack:

Category	Tools/Libraries
Deep learning	PyTorch + timm
Vision backbone	ConvNeXt-Tiny
Augmentation	Torchvision.transforms
Geospatial	GeoPandas + Shapely
Platform	Google Colab + Drive
Visualization	matplotlib + mpatches
Output format	GeoPackages(.gpkg)

Methodology (Approach 1) :



Approach-2:

Methodology:

1. Data Preparation in ArcGIS:

The provided drone orthophoto images and shapefiles were first imported into ArcGIS for preprocessing and analysis. Different infrastructure features such as buildings, roads, water bodies, and utility elements were identified and labeled using shapefiles.

2. Feature Labeling:

Each infrastructure feature was assigned a specific color code in ArcGIS to make visualization easier. Based on these labeled layers, four main infrastructure classes were selected for the model – Houses, Roads, Waterbodies & Electric Poles.

3. Image Export for Model Training:

After labeling, the processed geospatial data was exported in JPG and TIFF image formats. These images were then used to create the dataset required for machine learning training.

4. Image Tiling:

The large drone images were divided into smaller tiles of 256×256 pixels. This allows the system to analyze smaller image regions more effectively. A total of 2,424 tiles were generated from the dataset.

5. Binary Label Assignment:

For each tile, the presence of infrastructure features was detected using the RGB color values from the labeled images. Binary labels were assigned for each class:

- 1 → Feature present
- 0 → Feature absent

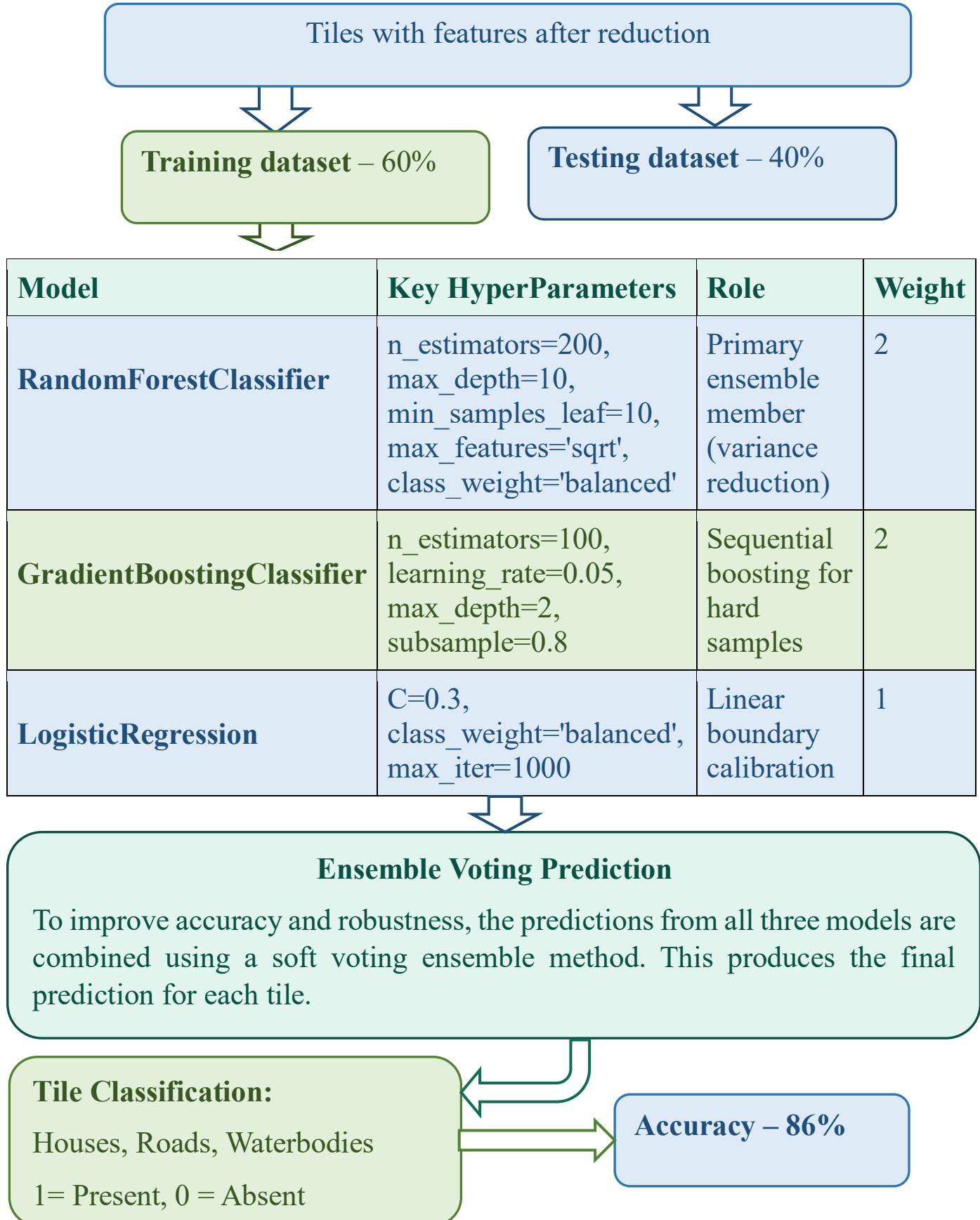
This created a multi-label dataset for training the model.

6. Feature Generation:



Category	Description
DWT(Haar Wavelets)	3-level discrete wavelet decomposition on grayscale. Energy and std of sub-bands at each level (LL, LH, HL, HH)
FFT Features	Log energy from 2D FFT; high-frequency ratio computed via radial frequency mask
Color Statistics	Mean, std, skewness per channel across BGR, RGB, HSV, LAB color spaces
Texture (GLCM)	Gray-Level Co-occurrence Matrix: contrast, dissimilarity, homogeneity, energy, correlation at 4 angles
Histogram Features	Per-channel histograms (8 bins) for grayscale, R, G, B; normalized bin counts
Edge / Gradient	Sobel gradients, Canny edge density, Laplacian variance (blur detection)
Shape / Structural	Contour count, area stats (mean, max, std), corner density (Harris/Shi-Tomasi)
Spectral Indices	NDVI proxy $(R-G)/(R+G+\epsilon)$, road/water/soil spectral ratios from RGB bands
Fourier Directional	Directional frequency energy (horizontal, vertical, diagonal) from FFT spectrum

7. Model:



Expected Impact:

1. Social Impact:

Faster rural infrastructure mapping reduces survey time from months to days, enabling quicker decision-making and improved governance at the local level.

2. Economic Impact:

Reduces dependency on manual surveys and GIS specialists, significantly lowering operational costs and making large-scale mapping more affordable.

3. Environmental Impact:

Accurate detection of ponds and land features supports water resource management and environmental monitoring across rural regions.

4. Infrastructure Impact:

Provides reliable mapping of houses, roads, and electric poles, helping authorities plan and manage rural infrastructure more effectively.

Required Resources:

Component	Specification
Hardware	NVIDIA A100/V100 GPU or Google Colab Pro+ for efficient training and inference
Data	SVAMITVA drone orthophotos with 2400+ labeled tiles
Storage	Minimum 50 GB cloud storage (Google Drive) for images, model weights, and outputs
Software	Python 3.12, PyTorch, timm, GeoPandas, Shapely, OpenCV (all open-source)
GIS Tools	QGIS / ArcGIS for validation

Scalability and sustainability:

Scalability:

- Processes any village image automatically using a tile-based approach
- Handles images of any resolution without retraining (only re-tiling required)
- Easily extendable to new classes (e.g., wells, overhead tanks) through relabeling
- Capable of scaling across many villages in India without architectural changes

Sustainability:

- Built using a 100% open-source stack → no licensing cost
- Model performance improves with additional training data over time
- Outputs in GPKG format, compatible with QGIS, ArcGIS, and other GIS tools
- Can be deployed as a REST API for integration with platforms

Current Stage of Development:

The project is at the **Prototype stage**, with an AI-based ConvNeXt-Tiny model that detects houses, roads, ponds, and electric poles from drone orthophotos using a tile-based approach, achieving ~91% accuracy.

While the initial scope included rooftop types, transformers, and utilities, the current system focuses on core features and provides an end-to-end pipeline with GIS outputs, with scope for future extension.

Conclusion:

This project demonstrates an effective AI-based approach for automated geospatial mapping of rural infrastructure using drone orthophotos. By leveraging a tile-based ConvNeXt-Tiny model, the system successfully detects key features such as houses, roads, ponds, and electric poles with an accuracy of approximately 91%. The developed pipeline efficiently handles large-scale images and generates GIS-ready outputs, reducing the need for manual surveys. While the current implementation focuses on core features, the system is scalable and can be extended to additional infrastructure elements. Overall, the solution provides a reliable, cost-effective, and practical approach for supporting data-driven rural planning and development.

Future Scope:

- Extend the model to detect additional features such as rooftop types, transformers, wells, and other utilities
- Improve accuracy by using larger and more diverse datasets
- Integrate the system with real-time drone data pipelines
- Deploy as a web application or REST API for government platforms like SVAMITVA
- Enhance geospatial outputs with higher precision and better visualization tools
- Explore advanced models for better performance on complex aerial imagery

Supporting Materials:

For Approach 1:

Material	Access links
Output(.gpkg file)	output(gpkg files) - Google Drive
Images	images - Google Drive
Metrics	Metrics - Google Drive
Github	GitHub - Kranthi1611/Geospatial_convNeXt-Tiny · GitHub

For Approach 2:

Material	Acaess links
Output(gpkg & csv)	outputs - Google Drive
Metrics	Accuracy Metrics - Google Drive
Feature extraction	Feature Extraction - Google Drive
Github	GitHub - mayirichennakesavarao/SVAMITVA-Geospatial-Intelligence · GitHub

Accuracy metrics:

Approach 1:

Combined Accuracy: 91.17161631584167
Prediction shape: torch.Size([606, 4])

Approach 2:

Houses	Roads	Ponds	Electric poles
89.4% test acc · F1 0.894	75.1% test acc · F1 0.749	94.5% test acc · F1 0.938	88.8% test acc · F1 0.883

Combined accuracy : 87%